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Economic assessment of hydropower capacity expansions and multi-market participation using a multi-stage stochastic framework

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***Abstract*—The growing need for flexibility in modern power systems increases the value of hydropower in capacity markets. This paper presents a stochastic optimization framework for quantifying the benefits of capacity market participation and evaluating hydropower capacity upgrades. Spot price outcomes for the 2030 electricity market have been simulated across multiple historical weather years and clustered using the K-means method. Uncertainty in energy prices and inflow is represented through a scenario tree constructed using Markov chain logic. Stochastic Dual Dynamic Programming has been applied to generate cuts embedded in a simulation model for statistical analysis. A case study of a NO2 hydropower plant indicates that combining energy and capacity market participation with a 75% capacity upgrade could enhance revenues by nearly 50% compared to energy-only participation. Future work could incorporate stochastic modeling of capacity market prices, reserve activation and market saturation, and tax-related considerations for hydropower producers.**

***Index Terms*— Ancillary service markets, Capacity upgrades, Hydropower scheduling, Multi-market operation, Stochastic dual dynamic programming.**

Sets

$\mathcal{W}$	Set of weeks in the planning period
Ω	Set of spot price–inflow scenarios, indexed by ω
$\mathcal{T}$	Set of time periods with hourly resolution
$\mathcal{E}$	Set of energy markets (DA, ID Sell, ID Buy)
$\mathcal{C}$	Set of capacity markets (aFRR Up, aFRR Down, FCR-N, FCR-D)
$\mathcal{C}^D$	Subset of down-regulating capacity markets, $\mathcal{C}^D \subseteq \mathcal{C}$
$\mathcal{C}^U$	Subset of up-regulating capacity markets, $\mathcal{C}^U \subseteq \mathcal{C}$
$\mathcal{S}^{PQ}$	Index set of points used in the linearized PQ-curve

Parameters

$P_{e,w,t,\omega}$	Price in market $e \in \mathcal{E}$ in week $w \in \mathcal{W}$ at time $t \in \mathcal{T}$ in scenario $\omega \in \Omega$ [EUR/MWh]
$P_{c,w,t}$	Capacity price in market $c \in \mathcal{C}$ in week $w \in \mathcal{W}$ at time $t \in \mathcal{T}$ [EUR/MW]
$\overline{P}_i^{hyd}$	Discretized hydraulic power at PQ-point $i \in \mathcal{S}^{PQ}$ [MWh/h]
$\overline{P}_i^{net}$	Net electrical power delivered to the grid at PQ-point $i \in \mathcal{S}^{PQ}$ [MWh/h]
M_i	Slope between consecutive PQ-points i and $i+1$, $\forall i \in \mathcal{S}^{PQ} \setminus \{\max(\mathcal{S}^{PQ})\}$
$I_{w,t,\omega}$	Reservoir inflow at time $t \in \mathcal{T}$ in week $w \in W$ in scenario $\omega \in \Omega$ [GWh/h]
$C^{\max}$	Maximum production capacity [MW]
$C^{\min}$	Minimum production capacity [MW]
$S^{\max}$	Maximum reservoir storage capacity [GWh]
$S^{\min}$	Minimum reservoir storage capacity [GWh]
δ^{up}	Maximum energy for up-regulation [GWh]
δ^{down}	Maximum energy for down-regulation [GWh]
$V_c^{\max}$	Maximum capacity available in market $c \in \mathcal{C}$ [MW]
$V_{e,t}^{\max}$	Maximum energy volume in Intraday market at time $t \in \mathcal{T}$ [MWh]
S^0	Initial reservoir storage level [GWh]
A_e	Market share in intraday market $e \in \mathcal{E} \setminus \{\text{DA}\}$ [–]
L^{em}	Cost of using emergency water [EUR/GWh]

Decision Variables

$y_{c,t,\omega}$	Capacity traded in market $c \in \mathcal{C}$ at time $t \in \mathcal{T}$ in scenario $\omega \in \Omega$ [MW]
$x_{e,t,\omega}$	Energy traded in market $e \in \mathcal{E}$ at time $t \in \mathcal{T}$ in scenario $\omega \in \Omega$ [MWh]
$s_{w,t,\omega}$	Reservoiar storage in week $w \in \mathcal{W}$ at time $t \in \mathcal{T}$ in scenario $\omega \in \Omega$ [GWh]
$p_{w,t,\omega}^{hyd}$	Extracted hydraulic power from reservoiar in week $w \in \mathcal{W}$ at time $t \in \mathcal{T}$ in scenario $\omega \in \Omega$ [MWh/h]
$h_{w,t,\omega}^{em}$	Emergency water needed in week $w \in \mathcal{W}$ at time $t \in \mathcal{T}$ in scenario $\omega \in \Omega$ [GWh]

This work was supported by the Research Council of Norway through the FME NTRANS - Norwegian Centre for Energy Transition Strategies (grant no. 296205). We also acknowledge the iDesignRES project which has received funding from the Horizon Europe research and innovation program under Grant Agreement No. 101095849.

I. Introduction

A. Motivation

Over the past decade, large-scale deployment of Variable Renewable Energy Sources (VRES) and the decommissioning of coal- and nuclear-based generation have altered European power-system dynamics [1]. Simultaneously, Norway has become more tightly interconnected with continental Europe through expanded cross-border transmission, making the Nordic system more exposed to the same developments [2]. These trends can reduce revenues from traditional energy-only trading, while reduced synchronous inertia and more volatile net-load patterns increase the value of flexible balancing services [3]. Reservoir-based hydropower possesses several

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characteristics well suited to provide such services, particularly when flexibility is required over extended periods.

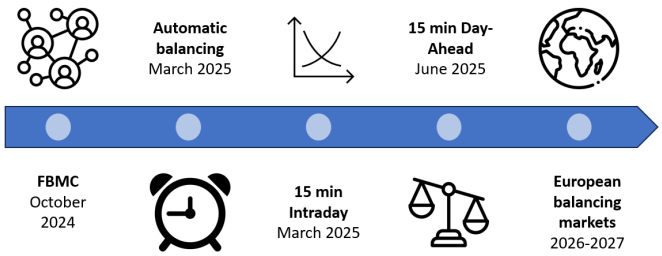


Fig. 1: Market changes in the European power system [4]

Concurrently with the changing generation mix, European power markets have undergone substantial reforms (see Fig. 1). Prices for balancing- and capacity-related products have increased in several markets, contributing to higher system costs borne by grid operators and market actors [4]. This development strengthens the investment case for flexibility-oriented upgrades and challenges traditional approaches to calculating *water values* [5].

In January 2026, Tussa Group approved a billion-NOK investment for a 90 MW capacity upgrade of an existing hydropower facility in western Norway [6]. As Norwegian regulatory frameworks largely protect unaltered watercourses while aging assets increasingly require reinvestment and rearmament [7], the Tussa II project exemplifies a broader strategic shift among Norwegian hydropower producers. Rather than developing new greenfield plants, capacity expansion and modernization of existing facilities are increasingly pursued to meet the requirements of tomorrow's power system. Consequently, models that capture multi-market participation and quantify the profitability of expansion measures under realistic market conditions are increasingly important.

B. Literature Study

Hydropower scheduling is inherently stochastic, as uncertain inflows and market prices interact with reservoir storage to create intertemporal trade-offs in production. The sequential decision structure and the intertemporal value of stored water motivate decomposition strategies. While Stochastic Dynamic Programming (SDP) captures this structure naturally, its practical application is limited by the curse of dimensionality [8]. Consequently, Stochastic Dual Dynamic Programming (SDDP) has become the state-of-the-art approach for medium term hydropower scheduling (MTHS) in hydro-dominated systems [9]. Pereira and Pinto [8] applied SDDP to a large-scale MTHS problem and demonstrated rapid convergence, establishing the methodological foundation for research on stochastic hydropower scheduling.

Parts of the research have focused on methodological extensions of the SDDP framework. To account for risk preferences, Shapiro et al. [10] introduced a worst-case expectation approach and showed that risk-averse formulations are less sensitive to demand shocks than risk-neutral models. Their analysis centered on robustness in aggregated systems. Helseth and Mo [11] improved the link between aggregated and detailed hydropower representations by introducing feasibility cuts into the framework. This extension provided more precise water value estimates, albeit at the expense of increased computational time. Similarly, Hjelmeland et al. [12] introduced strengthened Benders cuts and found improved policy quality, though at roughly five times the computational cost. While these studies have improved the framework, none of them quantify the economic value of coordinated participation across multiple electricity markets.

Several studies have incorporated multi-market participation into the SDDP framework. Using a hybrid SDP/SDDP approach to co-optimize energy and capacity sales in Norway, Helseth et al. [13] found that the multi-market extension had only a minor impact on computational burden, although simplifying assumptions may have led to an overestimation of achievable profits. Extending this framework, Helseth et al. [9] incorporated stochastic reserve prices and sequential market clearing, reporting an approximate 1% profit increase and limited impact on scheduling decisions. Pérez-Díaz et al. [14] conducted a similar study in Spain and reported a 0.12% profit increase. Their main contribution was modeling the hydropower producer as a price maker in the capacity market, which reduced spillage under strategic bidding behavior. While these studies report increased profits from participation in capacity markets, they do not evaluate these gains in the context of structural changes such as capacity upgrades.

Research from the investment perspective is limited, but Brekke et al. [15] studied a hypothetical rotational energy market and found revenues insufficient to justify investment. Although linking market design to investment incentives, the analysis was conducted during a period of systematically lower capacity prices and excluded capacity upgrade decisions.

Overall, existing literature demonstrates that SDDP-based models can incorporate multiple markets, risk preferences, and methodological refinements. However, limited attention has been given to quantifying the revenue potential of multi-market participation in the context of structural capacity changes within a stochastic framework.

C. Contribution

Motivated by the changing role of hydropower in modern power systems and the limited literature linking multi-market operation to investment decisions, this paper contributes to the field in two ways. First, it develops a stochastic MTHS model that explicitly represents participation in multiple energy and capacity markets. Lastly, the framework is used to evaluate whether the additional value from capacity market participation can justify capacity upgrades of hydropower plants.

The study addresses the following research questions:

- Can the additional value derived from capacity market participation justify a capacity upgrade of a hydropower plant?

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- How do various capacity upgrade configurations of a hydropower plant influence revenues across energy and capacity markets?

The remainder of this paper is structured as follows. Section II describes the methodology, followed by the case study in Section III and results in Section IV. Section V concludes the paper.

II. Methodology

A. Model description

To address the research questions in I-C, an optimization model has been developed for a hydropower plant and solved using an SDDP algorithm to account for uncertainty in the spot price and the inflow. The algorithm is decomposed into weekly decision stages, where realizations of spot prices and inflow for the period are assumed to be known. Additionally, a simulation model has been applied to estimate the results. The framework overview is highlighted in II-A1, while the optimization problem solved in the SDDP-algorithm and the simulation model is presented in II-B.



Fig. 2: Overview of the modeling framework

1) Framework overview: Figure 2 illustrates the framework applied in this study. Price and inflow data are processed in a clustering procedure to generate representative weekly combinations of spot prices and inflows using K-means clustering ($K = 3$) [16]. Short-term uncertainty is incorporated by allowing multiple realizations of the stochastic variables conditional on the previous weeks outcome. This dependency structure is modeled using a Markov chain [17], where the transition probabilities are estimated from the frequency of transitions between scenarios, given the current state.

Further, the processed data and Markov chain transition probabilities are fed into the SDDP algorithm used to generate Benders cuts. The reservoir storage level serves as the state variable linking consecutive weeks. For simplicity, the algorithm is described in algorithm 1 using one forward path per iteration, though multiple paths can be used.

Lines (1–2) initialize empty cut sets, store valid lower bounds, set the stability indicator to 0, and begin the iteration over M iterations. Further, it performs a forward pass (3–8), simulating L=1 number of future paths from the initial reservoir and Markov regime probabilities. This includes updating and solving the stage model, storing results, and preparing data for the next stage. In the backward pass (9–24), cuts are generated from the last stage to the first by solving models for each possible current regime across all next regimes, retrieving dual values, and aggregating α- and β-values into a single cut added to the stage. A convergence criterion has been applied in this study, though it can be omitted to let the loop run to completion. At each iteration, the current lower bound (LB) is computed and added to the set of valid, penalty-free LBs. The mean LB is updated, and stability is checked. The iteration stops once the change in the mean LB falls below a threshold, ϵ, and two consecutive valid solutions are obtained.

Algorithm 1: SDDP with W number of weeks (stages) in the planning period

Input: Stages $\{\mathcal{T}_w\}_{w=0}^{W-1}$, initial reservoir S_0, Markov $(\pi_0, \{P_w\})$, max iterations M, number of forward paths L
Output: Cuts K_w for $w = 0..W-2$

1 **Initialization:** $K_w \leftarrow \emptyset$; $validLB \leftarrow []$; $stable \leftarrow 0$.;
2 **for** $m = 1$ **to** M **do**
3 **Forward pass (L=1 paths):** $S \leftarrow S_0$; draw scenario $\omega \sim \pi_0$.;
4 **for** $w = 0$ **to** $W-1$ **do**
5 Update $Model[w, \omega]$ with scenario data (P_e, P_c, I) and starting storage S; solve.;
6 Store $v_w \leftarrow s^{end}$, $obj_w \leftarrow OBJ$, $\theta_w \leftarrow \theta$, pen_w.;
7 $S \leftarrow v_w$; draw next scenario ω;
8 **end**
9 **Backward pass (cut generation): for** $W = W-2$ **to** $w = 0$ **do**
10 Identify pairs of (ω_{curr}, v_w) (current scenario and end storage in week w):
11 **foreach** (ω_{curr}, v_w) **do**
12 **for** $\omega_{next} \in \{0, 1, 2\}$ **do**
13 Update and solve $Model[w{+}1, \omega_{next}]$ with starting storage S = v_w.;
14 $Q_{\omega_{next}} \leftarrow OBJ$;
15 $\lambda_{\omega_{next}} \leftarrow$ dual(Equation 3).;
16 **end**
17 $\alpha_{\omega_{curr},w} \leftarrow \sum_{\omega_{next}} P_w[\omega_{curr}, \omega_{next}]\, Q_{\omega_{next}}$;
18 $\beta_{\omega_{curr},w} \leftarrow \sum_{\omega_{next}} P_w[\omega_{curr}, \omega_{next}]\, \lambda_{\omega_{next}}$;
19 **end**
20 Aggregate over possible scenarios in week w:
21 $\alpha_w \leftarrow \sum_{r_{curr}} \pi_{\omega_{curr},w}\, \alpha_{\omega_{curr},w}$;
22 $\beta_w \leftarrow \sum_{\omega_{curr}} \pi_{\omega_{curr},w}\, \beta_{\omega_{curr},w}$;
23 Add single cut to week w: $\theta_w \leq \alpha_w + \beta_w(s^{end} - v)$;
24 **end**
25 **end**

Upon convergence of the SDDP algorithm, the resulting cuts are obtained. During the execution of the algorithm, cuts generated in each iteration are incorporated as constraints within the SDDP framework. The final set of cuts is subsequently implemented as constraints in the simulation model with a predefined number of iterations to estimate the revenues.

After conducting the simulations, the Net Present Value (NPV) method found in Figure 5 is applied to evaluate the incremental cash flows generated by the various capacity expansion projects described in Section III.

2) Model assumptions: The proposed methodology relies on several simplifying assumptions. The hydropower plant is assumed to be sufficiently small to operate as a price taker within the relevant price zone, and similar assumptions are commonly adopted in the literature, as observed in Helseth et al. [9]. Capacity markets are assumed to remain unsaturated and modeled without activation, implying that revenues stem solely from capacity procurements. To avoid computational complexity, the model does not consider start/stop conditions.

Long-term uncertainty is represented through a weekly Markov chain, which may limit the persistence and severity of extreme price-inflow combinations. Prolonged periods of scarcity or flooding would likely exacerbate price conditions.

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B. Optimization problem

1) Objective function: For a given week and price-inflow realization, the optimization problem is described in (1) is solved. The weekly objective is to maximize immediate and future revenues across markets. To ensure feasibility in the SDDP algorithm, the model is allowed to purchase water at a high penalty cost during intermediate iterations. Since the uncertain variables are realised within the current week, the scenario index is omitted for simplicity.

$$\max z_w = \sum_{e\in E\setminus\{\text{ID Buy}\}}\sum_{t\in T} P_{e,w,t}\, x_{e,w,t} + \sum_{c\in C}\sum_{t\in T} P^{c}_{c,w,t}\, y_{c,w,t} - \sum_{t\in T} P_{\text{ID Buy},w,t}\, x_{\text{ID Buy},w,t} - L^{\text{em}}\cdot\sum_{t\in T} h^{\text{em}}_{w,t} + \theta_w \quad (1)$$

2) Reservoir boundary constraints:

$$s_{w,t_0} = S^0, \quad w = 0 \quad (2)$$

$$s_{w,t_0} = s^{\text{end}}_{w-1,t_{end}}, \quad \forall w = 1,\dots,W \quad (3)$$

$$s_{w,t_{end}} = S^0, \quad w = W \quad (4)$$

$$S^{min} + \delta^{up} \;\le\; s_{w,t} + h^{em}_{w,t} \;\le\; S^{max} - \delta^{down}, \quad \forall w \in W, t \in T \quad (5)$$

The predefined limits of the reservoir and the storage capacity requirements are described in (2)-(5). For each consecutive stage, the following constraints must be satisfied. Within a stage, no further linking variables are necessary, so the stage index is omitted for simplicity.

2) Constraints for each stage:

$$p_t = \sum_{e\in E\setminus\{\text{ID Buy}\}} x_{e,t} - x_{\text{ID Buy},t}, \quad \forall t \in T \quad (6)$$

$$C^{\min} \le p_t \le C^{\max}, \quad \forall t \in T \quad (7)$$

$$p_t \le m_i\left(p^{hyd}_t - \bar{P}_i\right) + b_i, \quad \forall t \in T,\ \forall i \in \mathcal{I} \quad (8)$$

$$s_t = s_{t-1} + I_t - \frac{1}{1000}\, p^{hyd}_t, \quad \forall t \in T \setminus \{t_0\} \quad (9)$$

$$\sum_{c\in C_d} y_{c,t} \le p_t - C^{\min}, \quad \forall t \in T \quad (10)$$

$$\sum_{c\in C_u} y_{c,t} \le C^{\max} - p_t, \quad \forall t \in T \quad (11)$$

$$y_{c,t} \le V^{\max}_{c,t}, \quad \forall c \in C,\ \forall t \in T \quad (12)$$

$$x_{e,t} \le V^{\max}_{e,t} A_e, \quad \forall e \in E \setminus \{\text{DA}\},\ \forall t \in T \quad (13)$$

In (6)-(8), the energy production is defined, which must remain within the predefined limits and not exceed the linearized PQ-curve. The reservoir balance is ensured in (9), linking the current reservoir level to the previous state and the production. Respecting the available margin for selling downward and upward regulation is ensured in (10)-(11) respectively. Finally, (12)-(13) define the maximum trading limits for selling capacity in the capacity markets and energy in the intraday market.

III. Case Study

The modeling framework was applied to a case study of a hydropower plant in NO2 to assess the revenue potential from multi-market participation, given different capacity upgrade configurations. The input data used in the case study are summarized in Table IV, where the supply limits for the capacity markets are case-specific. Since the case study focuses on a forward-looking case in which a potential capacity upgrade is expected to have a long lifetime, spot price data were based on THEMA Consulting Group's Power Market Outlook, using the power system in 2030 and more than 40 different weather years.

To assess the added value from multi-market participation and different levels of capacity upgrades, eight instances were run using the model framework described in II-B. Three capacity upgrade instances, along with a trivial instance of no upgrade, were examined with and without capacity market participation (CMP).

Across the different instances considered in this study, the SDDP algorithm was run for between 10 and 31 iterations to generate the cuts. One cut was added per week per iteration and subsequently incorporated into the simulation model for the corresponding instance configuration. In the simulation model, 100 iterations were performed for each instance, and the resulting revenues were averaged across all iterations.

IV. Results & Discussions

This section presents the case study results, including annual revenues from CMP and its seasonal distribution. These results are compared to annual revenues without CMP to assess the added economic value, followed by an NPV analysis.

A. Relative change in total revenue given CMP

TABLE I: Total revenue with and without CMP for various capacity upgrades

Instance	With CMP (mEUR)	Without CMP (mEUR)
0 MW	23.71 (+25.84%)	18.84 (100%)
30 MW	26.47 (+38.86%)	19.06 (100%)
60 MW	28.32 (+50.73%)	18.79 (100%)
90 MW	30.56 (+57.46%)	19.41 (100%)

Table I shows the relative change in total revenue when CMP is included in the model. The results indicate that total annual revenues increases substantially when capacity markets are accessible. However, the absolute revenue gain grows at a diminishing rate as the capacity upgrade increases, as illustrated by comparing the 90 MW upgrade with the trivial instance. Capacity market revenues account for about 30% of total revenue in the former case, compared to roughly 20% in the trivial instance. This suggests that larger capacity upgrades can be more beneficial, although they might affect market saturation.

Further, revenues from the energy markets appear largely unaffected by increased capacity, although minor differences are observed. These differences may be related to variation in

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the number of iterations required for convergence. Nevertheless, since reservoir storage constraints must be satisfied given the inflow conditions, additional capacity cannot necessarily be converted into higher energy sales.

B. Capacity market revenues spread for capacity upgrades

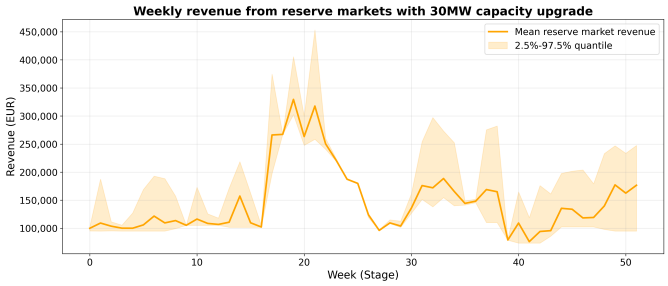


Fig. 3: Weekly revenue generated from CMP with 30MW upgrade.

Figure 3 suggests that there are large seasonal variations in revenues generated from participating in capacity markets. During late winter and early spring, the revenue potential is rather low compared to the spring flood occurring around April and May. It also shows that there are larger deviations from the mean revenue during the second half of the year.

TABLE II: Annual capacity revenue across upgrade instances with 95% CI.

Instance	Mean [mEUR]	2.5% [mEUR]	97.5% [mEUR]
0 MW upgrade	5.63	5.05 (-10.3%)	6.07 (+7.8%)
30 MW upgrade	7.68	7.12 (-7.3%)	8.26 (+7.6%)
60 MW upgrade	9.60	8.80 (-8.3%)	10.38 (+8.1%)
90 MW upgrade	11.46	10.66 (-7.0%)	12.34 (+7.7%)

Examining Table II reveals that annual mean revenue spans from 5.63 mEUR to 11.46 mEUR for non-upgrade and 90MW-upgrade, respectively. Based on the estimated 95% CI, the result may range from 10.3% below to 7.7% above the mean. Considering the magnitude of the revenue and the lifetime of a capacity upgrade, this indicates that an investor faces a prominent variability in expected revenues from capacity markets.

C. Magnitude on investment and OM-cost

TABLE III: NPV analysis of a 90 MW capacity upgrade using t=40 with CMP (without CMP)

r	NPV per MW [mEUR/MW]	NPV total [mEUR]
3%	1.76 (0.15)	158.25 (13.05)
7%	1.01 (0.08)	91.27 (7.53)
10%	0.74 (0.06)	66.95 (5.52)

Table III demonstrates the marginal NPV of the revenue over the project lifetime per MW of added capacity with and without CMP. For an investor, this value represents the maximum investment cost that can be justified when accounting for both the upfront capital expenditure and the discounted operation and maintenance (O&M) costs related to the capacity upgrade.

Depending on whether a conservative discount rate of 10% or a more industry-relevant rate of 6% is applied [18], the results indicate that approximately 50–75% of the reported investment costs for the Tussa II project [6] could be covered by the additional revenues from the capacity upgrade. By comparison, the NPV without CMP covers only 4–5.5% of the investment cost. Although the increased revenue from a 90 MW capacity upgrade alone may not fully cover the associated investment and O&M costs, the results highlight that participation in capacity markets provides substantially stronger investment incentives.

Apart from the chosen discount rate, several other factors may also affect the results. Revenue streams may materialize later in the project's lifetime due to long planning and construction lead times, which can reduce the overall NPV. Conversely, the model may also underestimate some of the revenue potential, as the mFRR capacity and energy activation market is not included. Furthermore, deterministic capacity market prices from 2024 have been used, and future prices may deviate from these levels in either direction, which motivates the need for modeling them stochastically. Klaeboe et al. [19] found no significant information between spot prices and balancing market prices, a finding that was later supported in the work by Backe et al. [20]. Hence, incorporating stochastic capacity market prices would likely require separate clustering and transition probabilities, thereby increasing model complexity. To mitigate the associated computational burden, a coarser temporal resolution could be applied in the SDDP algorithm.

V. Conclusion

This paper develops an SDDP framework to represent uncertainty in energy prices and inflow. Market simulations for 2030 across multiple weather years are used to generate correlated price-inflow trajectories, and the resulting SDDP cuts are embedded in a simulation model to assess the incremental revenue impacts of capacity upgrades and market participation strategies for a hydropower plant in NO2.

Results from the case study indicate that the additional economic value of a capacity upgrade, even when including CMP, is not sufficient to make the investment commercially viable under industry-specific discount rates. However, the inclusion of CMP yields an NPV that is about twelve times higher than without CMP, highlighting the importance of including capacity markets when evaluating such investment decisions. Given that hydropower producers have limited opportunities for greenfield projects, the additional value from CMP identified in this study is particularly relevant. The distribution of revenues across markets also suggests diminishing marginal returns from larger capacity upgrades when capacity markets are included. When capacity markets are excluded, revenues remain largely stable due to limited water availability for additional production.

Despite these results, limitations in market data representation remain. Future work should address stochastic capacity prices, reserve activation and saturation, mFRR markets, and tax considerations to improve realism and decision-support relevance.

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APPENDIX

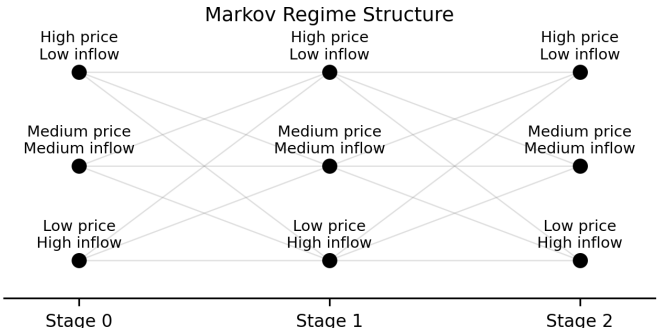


Fig. 4: Illustration of scenario tree propagation

TABLE IV: Summary of Case Study Input Data

Parameter	**Value**
Production limits	$[C^{min} = 15.504MW, C^{max} = 79.04MW]$
Reservoir storage capacity	[0 GWh, 101.2 GWh]
Intraday market share limit	10% of hourly volume
aFRR Up supply limit	$30\,\text{MW} \cdot \frac{\Delta C^{max}}{C^{max}}$
aFRR Down supply limit	$30\,\text{MW} \cdot \frac{\Delta C^{max}}{C^{max}}$
FCR-N supply limit	$8\,\text{MW} \cdot \frac{\Delta C^{max}}{C^{max}}$
FCR-D supply limit	$32\,\text{MW} \cdot \frac{\Delta C^{max}}{C^{max}}$
Capacity upgrade instances	0 MW, 30 MW, 60 MW, 90 MW
Capacity market prices	NO2 (2024)
Intraday prices	NO2 (2024)
Spot prices	Simulated 2030 outlook (hydro years 1980 - 2024)
Inflow data	Historical series (1980 - 2024), scaled to plant size
Discount rates (NPV)	3%, 7%, 10%
Planning resolution	Weekly stages, hourly operation

Fig. 5: NPV method used in the case study

r: discount rate [3%, 7%, 10%]
t: project lifetime [40 years]
CF: constant annual cash flow throughout the project lifetime
$A = \frac{r(1+r)^t}{(1+r)^t - 1}$
$NPV = \frac{CF}{A}$

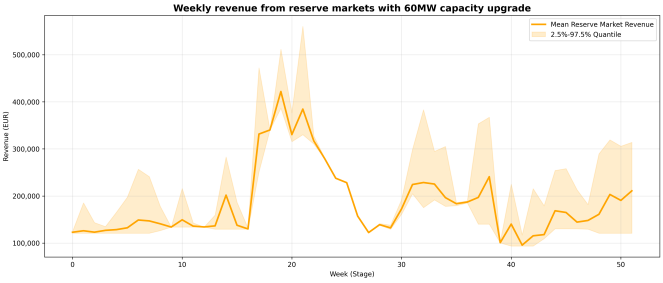


Fig. 6: Weekly revenue generated from CMP with 60MW upgrade.

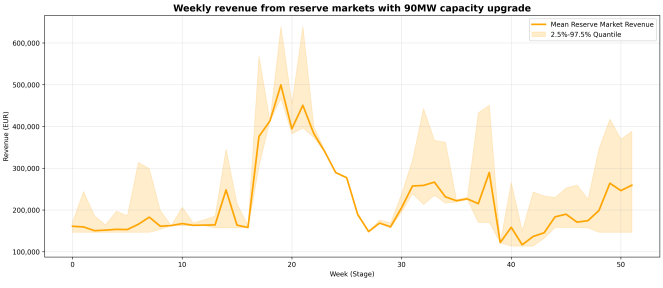


Fig. 7: Weekly revenue generated from CMP with 90MW upgrade.